

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married	
	Yes	No
Bachelors or higher	187	113
Some college	162	138
High school or less	149	151

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

		O	E	$\frac{(O-E)^2}{E}$
B	Yes	187	166	2.657
	No	113	134	3.291
C	Yes	162	166	0.096
	No	138	134	0.119
H	Yes	149	166	0.741
	No	151	134	2.157

$$\chi^2 = 10.061$$

$$(187 - 166)^2 = 498$$

$$\frac{498}{900} \times 300 = 166$$

$$(113 - 134)^2 = 492$$

$$\frac{492}{900} \times 300 = 134$$

$$\chi^2 = 10.061$$

$$\chi^2_{STAT} \sim \chi^2_{0.01, (3-1)(2-1)} = \chi^2_{0.01, 2}$$

$$\chi^2_{0.01, 2} = 9.210$$

$$RR: \{ \chi^2 \mid \chi^2 > 9.210 \}$$

$$10.061 > 9.210 \Rightarrow \text{Reject } H_0$$

We have sufficient evidence to say that marriage rates differ significantly among different educational level